

Internship Take-Home Assignment

Role: Software / AI Engineering Intern **Time:** 2 days **Deliverable:** Working code + extracted images + short README

Overview

You are given one PDF, **INPUT_images_and_questions.pdf**, containing **10 real photographs**. Each photo shows a group of objects (flowers, animals, vehicles, produce, etc.) and a question printed below it. Every question asks you to **extract one specific object** from that photo — for example, “Crop out only the bunch of yellow tulips and save it as 01_yellow_tulips.png.”

Your Task

Write a program that reads the input PDF and, for each of the 10 images, produces a cropped image file containing **only the requested object**. Your program should:

- Load the 10 photographs from the input PDF.
- For each photo, locate the object named in that question.
- Crop that object out and save it with the exact filename the question gives (e.g. **01_yellow_tulips.png**).
- Repeat until all 10 objects have been extracted and saved.

How you find the object is up to you — an object-detection model (YOLO, Grounding-DINO, SAM, a vision-LLM, etc.), classical colour/shape detection, or even manually-tuned coordinates are all acceptable. We care about a working, readable solution and clear reasoning.

What to Submit

- Your source code (git repo or zip).
- The 10 output images.
- A short README: how to run it, your approach, and any assumptions or limitations.

Tools & AI Assistance

You are **encouraged** to use any language, libraries, and AI-assisted development tools (ChatGPT, Claude, Copilot, Cursor, etc.). Using AI well is part of the job — just be ready to explain how your solution works and why you made your choices.

Evaluation

- **Correctness** — the right object is cropped for each image.
- **Code quality** — clear, organised, runnable.
- **Approach** — sensible method, honest handling of tricky cases.
- **Communication** — a README we can follow.

Deadline

Please submit within **2 days (48 hours)** of receiving this. If something is unclear, state your assumption in the README and proceed — reasonable assumptions are fine.

Good luck — we're looking forward to seeing how you think.